TechBiz Academy

MARYAM REHAN

CCNA Internship Report

Week 1 · Task 2 — IPv4 Addressing and Subnetting

Q1. Decimal to Binary Conversion

1. Decimal 192

192 = 128 + 64

Binary = 11000000

2. Decimal 168

168 = 128 + 32 + 8

Binary = 10101000

3. Decimal 10

10 = 8 + 2

Binary = 00001010

4. Decimal 255

255 = 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1

Binary = 11111111

5. Decimal 224

224 = 128 + 64 + 32

Binary = 11100000

6. Decimal 128

128 = 128

Binary = 10000000

7. Decimal 64

64 = 64

Binary = 01000000

8. Decimal 32

32 = 32

Binary = 00100000

9. Decimal 16

16 = 16

Binary = 00010000

10. Decimal 15

15 = 8 + 4 + 2 + 1

Binary = 00001111

Binary to Decimal Conversion

1. 11000000

128 + 64 = 192

2. 10101000

128 + 32 + 8 = 168

3. 00001010

8 + 2 = 10

4. 11111111

128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255

5. 11100000

128 + 64 + 32 = 224

6. 10000000

128 = 128

7. 01000000

64 = 64

8. 00100000

32 = 32

9. 00010000

16 = 16

10. 00001111

8 + 4 + 2 + 1 = 15

Q2. Networking Terms

1. **Network Address:** The first IP address of a subnet that identifies the network. Example: 192.168.10.0

2. **Broadcast Address:** The last IP address of a subnet used to communicate with all devices in that network. Example: 192.168.10.63

3. **Usable Host Range:** The IP addresses available for assigning to computers and network devices. Example: 192.168.10.1 – 192.168.10.62

4. **Subnet Mask:** Identifies the network and host portions of an IP address. Example: 255.255.255.0

Q3. VLSM Subnetting

Given Network: 192.168.10.0/24

Required Hosts: 50, 25, 12, 2, 2

Step 1: Subnet Calculations

50 Hosts: $2^6 - 2 = 62$ usable hosts, Prefix = /26

25 Hosts: $2^5 - 2 = 30$ usable hosts, Prefix = /27

12 Hosts: $2^4 - 2 = 14$ usable hosts, Prefix = /28

2 Hosts: $2^2 - 2 = 2$ usable hosts, Prefix = /30

2 Hosts: $2^2 - 2 = 2$ usable hosts, Prefix = /30

Step 2: VLSM Allocation

Subnet 1 — 50 Hosts

Network Address: 192.168.10.0/26

Subnet Mask: 255.255.255.192

Usable Host Range: 192.168.10.1 – 192.168.10.62

Broadcast Address: 192.168.10.63

Subnet 2 — 25 Hosts

Network Address: 192.168.10.64/27

Subnet Mask: 255.255.255.224

Usable Host Range: 192.168.10.65 – 192.168.10.94

Broadcast Address: 192.168.10.95

Subnet 3 — 12 Hosts

Network Address: 192.168.10.96/28

Subnet Mask: 255.255.255.240

Usable Host Range: 192.168.10.97 – 192.168.10.110

Broadcast Address: 192.168.10.111

Subnet 4 — 2 Hosts

Network Address: 192.168.10.112/30

Subnet Mask: 255.255.255.252

Usable Host Range: 192.168.10.113 – 192.168.10.114

Broadcast Address: 192.168.10.115

Subnet 5 — 2 Hosts

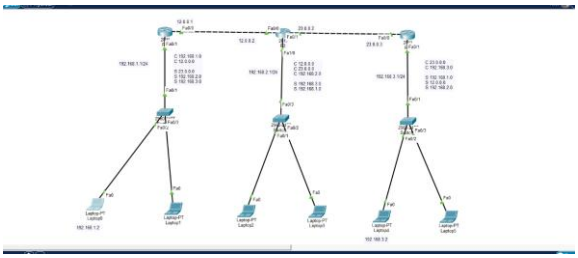
Network Address: 192.168.10.116/30

Subnet Mask: 255.255.255.252

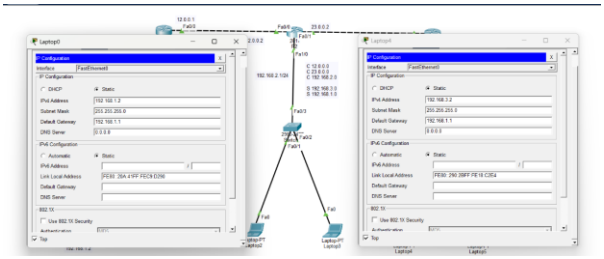
Usable Host Range: 192.168.10.117 – 192.168.10.118

Broadcast Address: 192.168.10.119

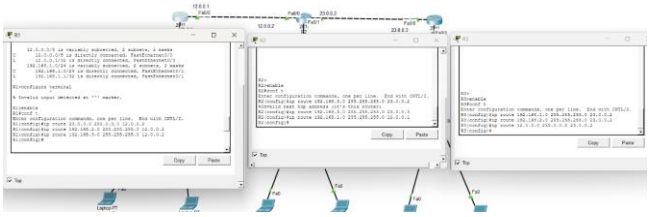
Static Routing



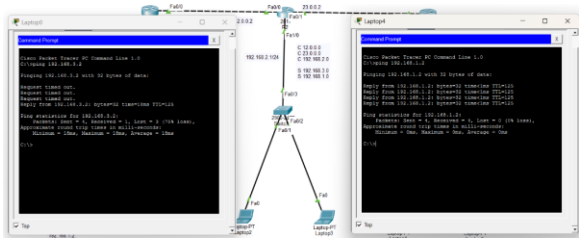
IP Addressing



Static Routing



Ping



1. Objective

To perform IP addressing and static routing in Cisco Packet Tracer and establish communication between different networks.

2. Explanation

I performed a networking practical in Cisco Packet Tracer using routers, switches and PCs. I assigned IP addresses and subnet masks to the devices, configured static routing on routers and tested network connectivity using ping commands.

3. Conclusion

The practical was completed successfully. I learned how to configure IP addresses, implement static routing and verify communication between network devices using Cisco Packet Tracer.